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The Lesser Apes

Evolutionary and Behavioural Biology

HOLGER PREUSCHOFT
DAVID J. CHIVERS
WARREN Y. BROCKELMAN
NORMAN CREEL

Editors

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1984

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24. Feeding and Ranging in Gibbons: A Summary

Introduction

In this section ecological data are presented on eight of the nine gibbon species, the only omission being the endangered concolor gibbon of Indo-China. Information is most detailed on the siamang, lar, agile and Kloss gibbons, and sketchiest on the hoolock, with important recent studies on the pileated, moloch and Müller's gibbons. There are nine main study sites represented – in Assam, Bangladesh, Thailand, Peninsular Malaysia, the Mentawai Islands, Java and Kalimantan (Fig. 24.1) – in the two climax forest formations: the deciduous monsoon forests of continental Asia to the Isthmus of Kra, and the evergreen rain forests of the Sunda Shelf (Whitmore, 1975).

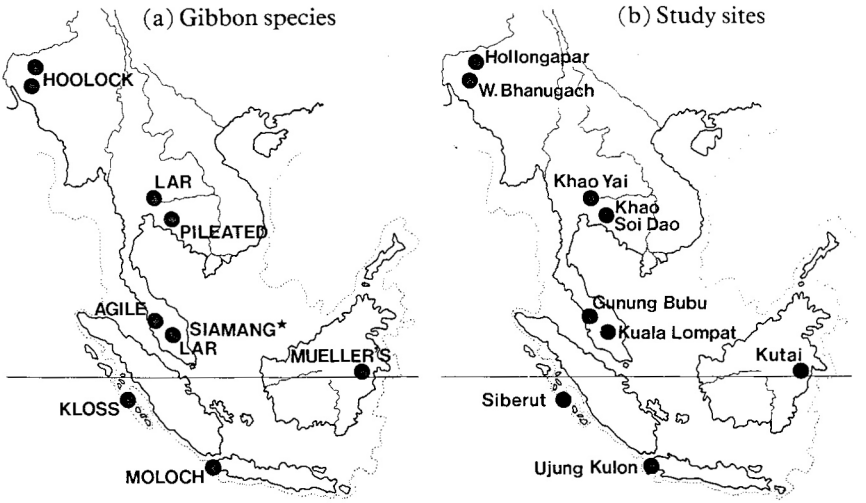


Figure 24.1. Distribution of (a) gibbon species and (b) study sites in south-east Asia.

Population Structure

The social organization of gibbons into monogamous family groups has been well known since Carpenter's (1940) pioneering study. Subsequent studies of different taxa have confirmed this pattern; average group size invariably approximates the modal group size of four animals (Ellefson, 1974; Chivers, 1974, 1977b; Raemaekers, 1977; Gittins, 1979; Tilson, 1981; chapters in this section). Apart from the adult male and adult female there may be up to four young (infant, juvenile, subadult), but there are usually only two; since they are born singly at two to three-year intervals,

Table 24.1. A summary of gibbon ecology.

Species	Location/observers	Ranging: Density (gp/km²)	Ranging: Home range (ha)	Ranging: Territory (ha)	Ranging: Day range (m)	Ranging: Night trees	Feeding: min./day	Feeding: F+FL	L	A	Travel min./day	Active period (min.)	Feed time as % AP	FT/TT	Travel time as % AP	Biomass kg/km² of HR	Indices: Fruit/Leaves	Indices: Day range/Biomass	Indices: Day range/Home range	Indices: Day range/Fruit
Siamang	Kuala Lompat, W. Malaysia																			
	D. J. Chivers *et al.*	0.8	32 (42)[1]	(26)[1]	870	34+	342	44	48	8	101	624	55	3.4	16	97	0.9	9.0	27.2	5.8
	J. J. Raemaekers		47		740	70	312	44	43	15	138	618	50	2.3	22		1.0	7.6	15.7	5.4
Hoolock	Hollongapar, Assam, India																			
	R. L. Tilson	2.2	22	17				67	32	1							2.1			
	West Bhanugach, Bangladesh																			
	S. P. Gittins	<4.4	23				308	67	32	1							2.1			
Kloss	Siberut, Mentawai Is., Indonesia																			
	R. Tenaza and R. Tilson		7-12	2-3																
	A. J. Whitten	2.8	31-35	21	1514	52	151	72	2	25	121	637	24	1.2	19	43	24.0	35.2	44.5	13.9
Moloch	Ujung Kulon, Java, Indonesia																			
	M. Kappeler	2.7	17	16	1400	19	192	62	38	0.2	135	600	32	1.4	23	98 (42-57)[2]	1.6	14.3 (25-33)[2]	80.5 (35-47)[2]	11.8
Pileated	Khao Soi Dao, Thailand																			
	S. Srikosamatara	5.0	25 (36)[1]	(27)[1]	833		128	71	13	15	123	492	22	0.9	25	60	5.5	13.9	23.1	9.2
Müller's	Kutai, Kalimantan, Indonesia																			
	D. Leighton	3.2	36 (44)[1]	(39)[1]	850	117+	199	69	29	1.5	58	486	41	3.4	12	35	2.4	18.8	19.5	6.2
Agile	Gunung Bubu, W. Malaysia																			
	S. P. Gittins	4.3	29	22	1335	30	200	61	39	1	165	550	36	1.2	30	51	1.6	26.1	46.0	10.9
Lar	Kuala Lompat, W. Malaysia																			
	D. J. Chivers *et al.*	0.7	55		1670		217	61	30	9	166	516	42	1.3	32	29	2.0	57.6	30.4	12.7
	J. J. Raemaekers		57		1430	176	216	57	29	13	168	516	42	1.3	33		2.0	49.3	25.1	11.6
	Khao Yai, Thailand																			
	W. Y. Brockelman	<4.0																		
	J. J. Raemaekers *et al.*	6.5	16 (14)[1]	(11)												104				

[1] Values for one group, where sample mean available for home range size.

[2] Values from survey area.

there will be only one representative of each immature age class at any one time.

All data collected since Carpenter's study are summarized in Table 24.1, and displayed in Figs 24.2–8; all scores are derived from at least one annual cycle. Differences in methods between observers, reduced by collaboration, do not generally affect the comparability of the results, although some possible exceptions will be indicated.

In the north of their distribution there are usually at least 4 gibbon groups/km², while in the islands of the Sundra Shelf there tend to be only 3 (Fig. 24.2a). Where two species occur sympatrically, as in the Malay Peninsula, there may be only one group of each species in each km² (Raemaekers, Chapter 19). In disturbed habitats numbers may be increased or decreased according to the nature of the disruption, e.g. groups may be crowded together (without migration) by clearance of the surrounding forest, or thinned out by hunting.

Biomass varies from 30 to 100 kg/km² of home range (Fig. 24.2b) and, again, is surprisingly high in the northern, drier and more seasonal forests. Where two species occur in sympatry, the total gibbon biomass is highest (mainly because of the larger size of the siamang, which is always present in these cases), but the smaller gibbon (lar or agile) is usually at a lower density than elsewhere in their ranges.

Ranging Behaviour

The number of night sleeping sites used is very variable between species (Fig. 24.3a), presumably because of idiosyncrasies in habitat structure. Thus, over a year, lar and Müller's gibbons used more than 100 trees, whereas the moloch gibbon used only 19; and in one site the Kloss gibbon used only 2–3 trees, but 50 in another part of the same Reserve.

Day range varies between 1300 and 1700 m for most of the smaller rain forest gibbons, but, like the large siamang, the pileated and Müller's gibbons travel only 850 m daily on average (Fig. 24.3b).

Home ranges are largest in the centre of the gibbon distribution, in the

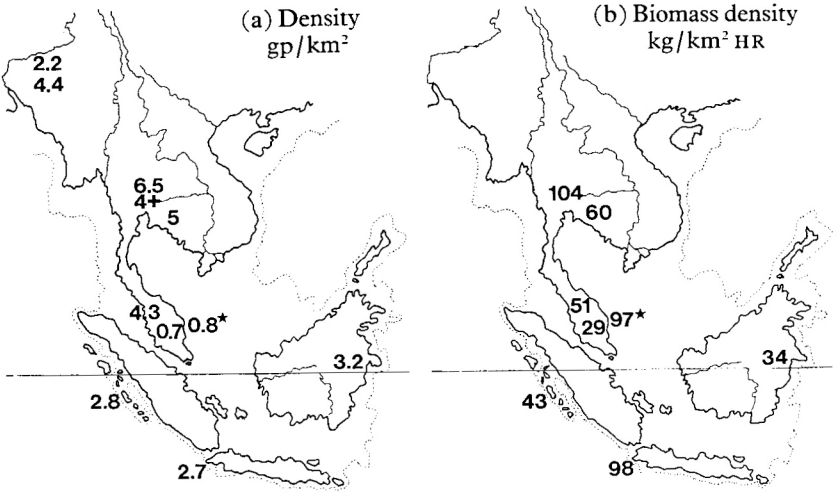


Figure 24.2. (a) Density and (b) biomass density of gibbons in south-east Asia.

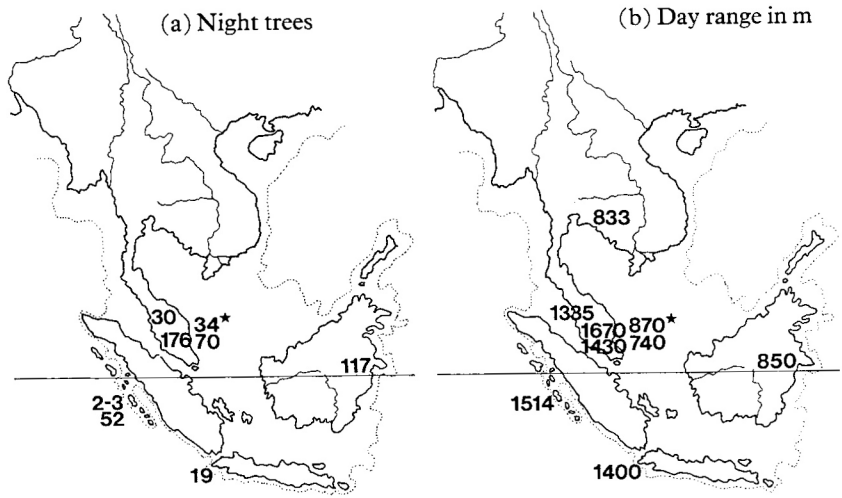


Figure 24.3. (a) Number of night trees and (b) mean day range of gibbon species.

Malay Peninsula: over 50 ha for the lar gibbon in sympatry with the siamang or 44 ha for the lar gibbon alone (Ellefson, 1974). Ranges are smallest at the periphery: 22 ha for the hoolock in Assam and Bangladesh and 17 ha for the moloch gibbon in Java (Fig. 24.4a). Raemaekers *et al.* (pers. comm.) have just reported home ranges of 10.5–21.4 ha for 8 groups of lar gibbon in the Khao Yai Park in Thailand, an average range of 15.5 ha, with 6.5 groups of

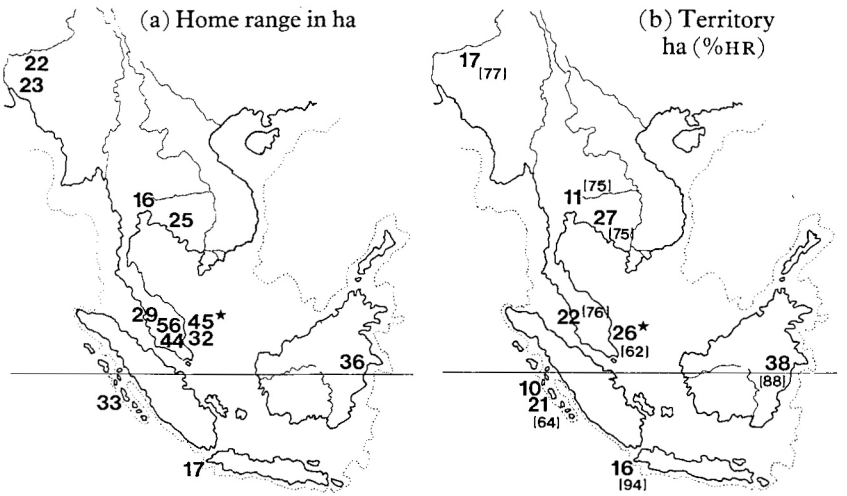


Figure 24.4. (a) Mean home range area and (b) mean territory size (and as percentage of home range, HR, in brackets) in gibbon species.

4.0 individuals/km². Mean home range size for the 13 sample localities is 34.2 ha (modal value 32 ha).

About 75 per cent of the home range is defended against neighbouring conspecifics for the exclusive use of the resident group; the territory is a smaller proportion of the home range in siamang and Kloss gibbons, and larger in moloch and Müller's gibbons (Fig. 24.4b). The actual sizes of these territories vary between 11 and 38 ha, with the largest and smallest towards the periphery of gibbon distribution and the modal scores (ca. 25 ha) in the centre.

Activity Patterns

Gibbons are unusual among tropical mammals in usually having a shorter, essentially unimodal, period of activity. Instead of being most active soon after dawn and before dusk, they maintain a high level of activity during the morning, which declines during the afternoon until they retire to sleeping positions several hours before dusk. The siamang, Kloss and moloch gibbons are active for ten hours or more daily, whereas pileated and Müller's gibbons are active for only about eight hours, with the lar and agile gibbons of the Malay Peninsula for around nine hours (Fig. 24.5a).

Between 22 and 50 per cent of the active period is spent feeding (Fig. 24.5b), and between 12 and 33 per cent travelling (Fig. 24.5c). The larger siamang spends the most time feeding (more than 5 hours daily), the Kloss and pileated gibbons the least (2–2.5 hours); this relates to the kinds of food ingested (see below). Otherwise, gibbon species score between 3 and 4 hours of feeding activity. Kloss and Müller's gibbons spend the least time travelling, lar and agile gibbons the most.

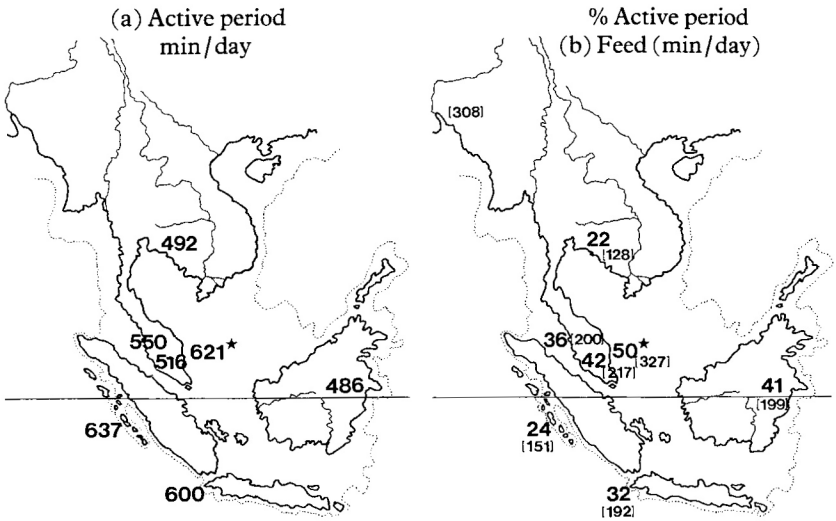
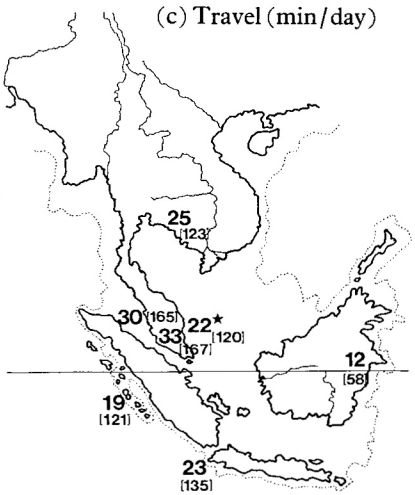


Figure 24.5. (a) Mean duration of active period daily and percentage of active period spent, (b) feeding and (c, *opposite*) travelling (with mean no. mins daily in brackets) by gibbon species.

Comparisons of feeding and ranging are hindered by differences in sampling techniques and in the behaviour of the animals themselves. More precise one-minute samples of travel, recording stationary spells during bouts, will underestimate the amount of time spent travelling around the home range, while ten-minute samples may overestimate it; not all observers have used five-minute samples of major activities. Social groups of some species feed mainly in discrete bouts, which aids scoring, whereas in others the group scatters and forages here and there as the group travels between large fruiting trees; this can lead to an underestimation of feeding, especially by some recording techniques. It is difficult to quantify the amounts of time spent feeding and moving during foraging bouts; apart from keeping in visual contact with the animals, much time is spent inspecting the vegetation for plant or animal food.

Diet

Gibbons have for long been described as frugivorous, and our studies confirm this. MacKinnon and MacKinnon (1978) and Raemaekers (1977) in particular have shown the importance of small, scattered fruit sources to gibbons; such fruiting trees or climbers are too small for the larger groups of monkeys (Raemaekers and Chivers, 1980). Between 57 and 72 per cent of feeding time is spent eating the reproductive parts of plants (fruit and flowers), except for the siamang which only spends 44 per cent of feeding time in this way. The lar and agile gibbons spend about 60 per cent of feeding time like this, the rest around 70 per cent for their annual average



(Fig. 24.6a). Young leaves are a significant component of the diet for most gibbons, especially the larger, more sedate siamang (43–48 per cent, annual averages), but the Kloss gibbon eats virtually none (Whitten, Chapter 20), and the pileated gibbon only 13 per cent (Fig. 24.6b). Invertebrates provide an important source of animal protein, especially for the Kloss gibbon (25 per cent of feeding time); however, outside the Malay Peninsula and Thailand invertebrate scores are very low (Fig. 24.6c).

The differences in diet relate to differences in habitat and food availability, and to the fact that protein can be obtained either from plants or animals. Low leaf intake seems to relate to forest on poor soils with plants that are better protected chemically (Whitten, 1980). Otherwise, the different species show a similar preference for fleshy, sugary fruit and figs.

Feeding and Ranging Comparisons

The ratio of feeding time to travel time is highest in the siamang and Müller's gibbons (3.4) and lowest in the pileated gibbon (0.9); otherwise, the ratios are between 1.2 and 1.4 (Fig. 24.7a). Time spent eating fruit is about double that for leaves, except for the siamang where it is equal, the pileated gibbon where it is five times greater, and the Kloss gibbon where it is twenty-four times more. The index of day range length to fruit-eating time is about 12 for all taxa, except for the siamang and Müller's gibbon where it is about 6 (Fig. 24.7b). Thus, the least frugivorous species, and one of the most frugivorous, obtain their fruit over half the distance covered by other species (although not all of this is spent searching for food).

The index of day range to biomass is very variable (Fig. 24.8a); the

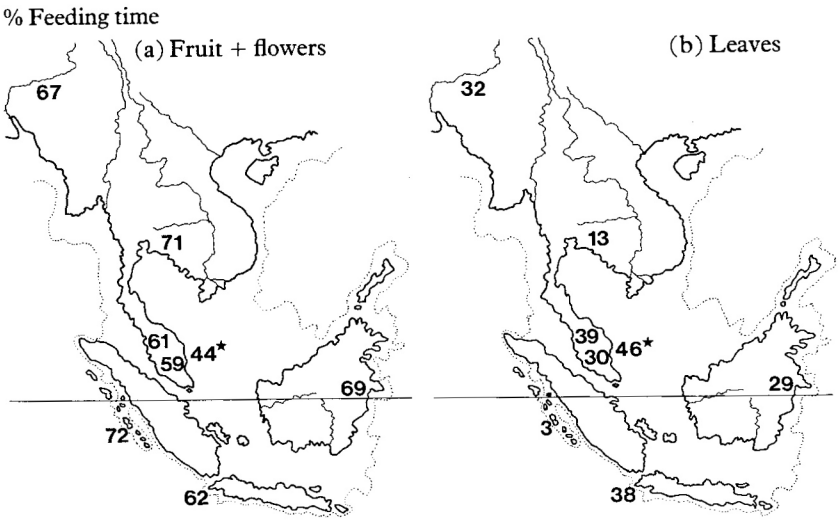


Figure 24.6. Percentage of feeding time spent on (a) fruit and flowers, (b) leaves and (c, *opposite*) animal matter by gibbon species.

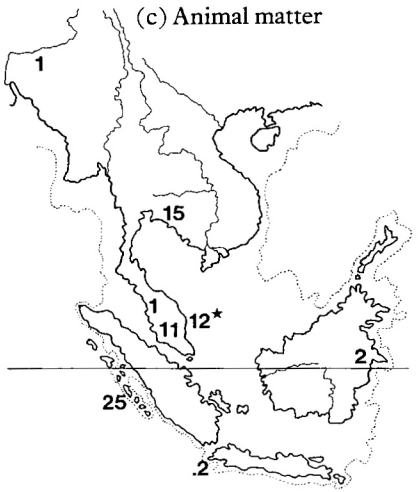
siamang has the lowest index of 8, the sympatric lar gibbon has the highest of 49, and the pileated, moloch and Müller's gibbons cluster between 14 and 19. The index of day range length to home range area tends to be higher in those species with smaller home ranges, varying between 20 and 46, except for the moloch gibbon at 81 (Fig. 24.8b).

Discussion

With the exceptions indicated, the lesser apes are characterized by living in monogamous family groups of about 4 individuals, with 2–4 groups/km^2 (or 40–60 kg/km^2). Groups travel about 1 500 m/day around a home range of 30–40 ha of which 75 per cent (25 ha) can be identified as a territory. They are active for 9–10 hours each day, feeding for 35 per cent of that time and travelling for 24 per cent. Apart from the siamang, 66 per cent of feeding time involves fruit, and, apart from the Kloss gibbon, 32 per cent concerns young foliage.

It seems that gibbon food preferences limit them to certain kinds of forest with some distinctive features of food supply. Despite the numerous field studies, we still do not understand the relationships between food supply and gibbon densities, or foraging strategies in the face of fluctuating food supplies. The evergreen rain forests are unparalleled in their complexity, in their high diversity and abundance of plant and animal species. A study in a marginal habitat (drier, more seasonal, less diverse) could help to remedy these deficiencies.

If food is limiting on density, reproductive plant parts are the key



factor (Table 24.2); they are scarce relative to vegetative parts, many cannot be eaten by gibbons (including several families comprising a substantial part of the forest biomass; others are too large or well-defended mechanically), and their preference for ripe fruit may be pre-empted by competitors which can overcome the chemical defences of unripe fruit. Their poor detoxification ability, in relation to a low-fibre diet, could be a key factor restricting gibbons to the more diverse and less seasonal forests, where toxic effects can be reduced by frequently changing foods (Vellayan, 1981). Gibbons prefer fruit with juicy pulps, maintaining a year-round intake; the structurally complex environment buffers the effects of seasonality, enabling animal-dispersed fruit to be available throughout the year (Smythe, 1970). In exploiting small, scattered fruit sources, gibbons come

Table 24.2. Relative abundance of and selection for different food classes. Siamang and lar gibbon, 12 months.

Food category	Group	Feeding opportunities provided	Feeding opportunities taken	Taken as % provided
Fig fruit	Siamang	26	10	39
	Lar	25	10	40
Other fruit	Siamang	71	7	10
	Lar	97	9	9
Flowers	Siamang	19	2	10
	Lar	15	2	13
Young leaves	Siamang	308	6	2
	Lar	273	7	3

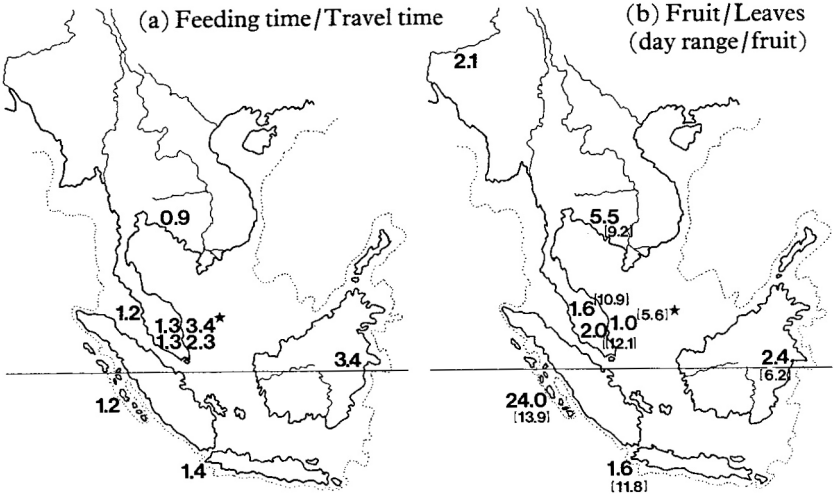


Figure 24.7. Ratios of (a) feeding time to travel time and (b) fruit to leaves (and day range to fruit-eating in brackets) in gibbon species.

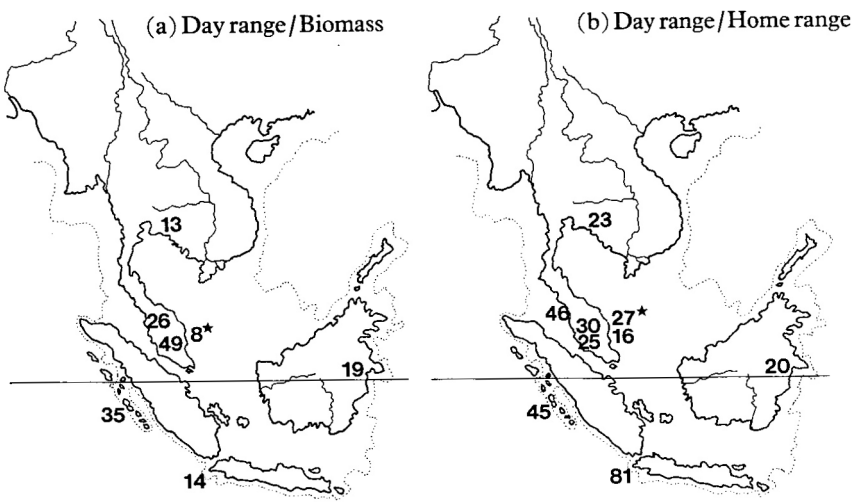


Figure 24.8. Ratios of (a) day range to biomass and (b) day range to home range in gibbon species.

into competition more with birds and squirrels than with other primates (Payne, 1979).

The year-round availability of fruit must be the key requirement restricting gibbons to forests that are otherwise characterized by high floristic diversity and only moderate seasonality in productivity. In one year at Kuala Lompat (West Malaysia) almost all tree species and individual

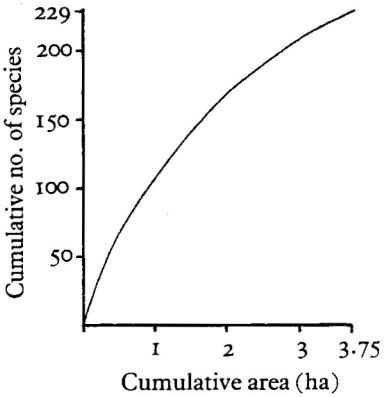


Figure 24.9. Tree species/area curve: increase in the number of species of trees exceeding 50 cm girth tallied as the area of the botanical sample is increased. The curve shown is the average of three almost identical plots obtained from three random orderings of all plots except P10. (From Raemaekers *et al.*, 1980.)

trees bore young leaves at some time; only 56 per cent of species and 31 per cent of individual trees flowered; and only 33 per cent of species and 18 per cent of individual trees fruited (sample of 910 trees over 50 cm girth at breast height). In that year 50 per cent of individual trees bore young green parts eaten by gibbons, but only 8 per cent bore fruit that gibbons eat.

Although gibbons occur over a wide geographical range, they actually inhabit only the moister forests therein. In Peninsular Malaysia, Marsh and Wilson (1981) found by multiple regression that tree family diversity 'explained' most of the variance in gibbon density between forest types. Kuala Lompat is an example of an extremely diverse gibbon habitat (Fig. 24.9); here, a primate social group occupying 60 ha has access to over 400 species of tree and liana (Raemaekers *et al.*, 1980). In Huay Kha Khaeng, Thailand, by contrast, lar gibbons thrive in a mosaic of dry evergreen, mixed deciduous and dry dipterocarp forests, and there is a dry season of several months. The gibbons do not enter the dry dipterocarp forest, which is of low species diversity and mostly burnt annually, but they do use the other two types, which are certainly of a lower diversity than the rain forests of the Sunda Shelf.

Their consistent intake of juicy fruit pulp is achieved despite some degree of seasonality, occurring even in lowland rain forest like that at Kuala Lompat (Fig. 24.10). The gibbons, which seldom destroy the seeds of fruit they eat and therefore disperse them, might moderate the seasonality among their fruit food species by selecting for asynchrony between species and individuals (Snow, 1965; Smythe, 1970). At Kuala Lompat there does seem to be some tendency towards asynchronous fruiting among sets of animal-dispersed species with morphologically similar fruit (Table

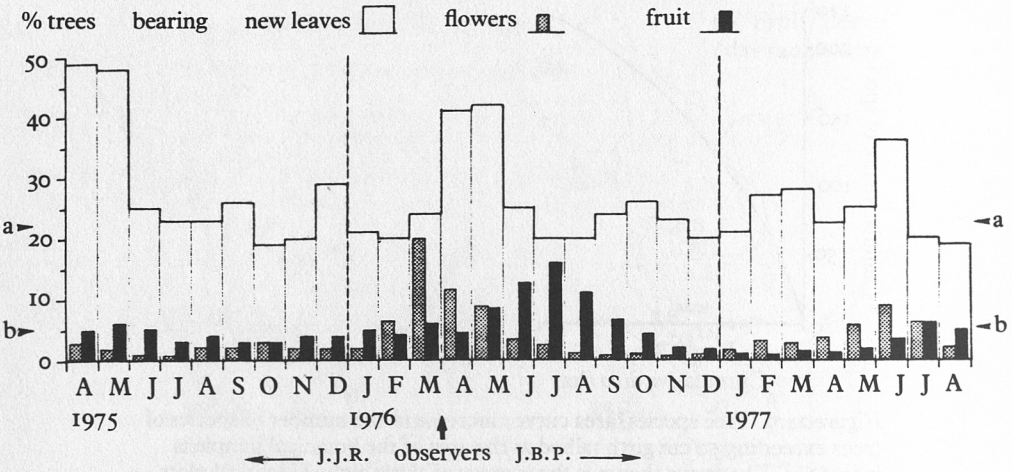


Figure 24.10. Percentage of trees with leaf flush, flowers and fruit in each month of the study of 900 trees at Kuala Lompat from 1975 to 1977. (a) marks level of mean for leaf flush from July 1975 to February 1976; (b) marks same from July 1976 to February 1977. (From Chivers and Raemaekers, 1980.)

24.3; D.S.R. Leighton, in prep.). Even so, large differences between months in the relative abundance of food types remain (Fig. 24.11). Any selective pressure that the gibbons might exert may be swamped by counter-pressure exerted by species, such as some monkeys and squirrels (not usually birds) that eat many of the same fruits but destroy many of their seeds.

We know little as yet about how gibbons cope with seasonal variations in food supply, since most field studies have not provided data on food supply as distinct from food consumption. Where this has been done, as at Kuala Lompat, siamang and lar gibbons were found to respond to temporary falls in supply by reducing day range and concentrating on a very few sources of high-energy fruit (Raemaekers, 1980). D.S.R. Leighton (in prep.) will soon produce a detailed analysis of the situation in Kutai, Kalimantan (Indonesian Borneo). How do gibbons operate in an environment like Huay Kha Khaeng? They certainly enter the mixed deciduous forest, even at the height of the dry season (J.J. Raemaekers, pers. comm.),

Table 24.3. Asynchronous fruit production by sets of tree species with morphologically similar fruits, Kuala Lompat 1976-77. N = number of trees observed with fruit. C = canopy levels in which fruit is borne: E, emergent; U, upper; M, middle; L, lower. L = length of fruit (mm). At Kuala Lompat, eaten by: gibbons (G), macaques (M), hornbills (HB), pigeons (P), barbets (B); other small birds, including bulbuls, flowerpeckers, leafbirds (SB) (all mainly seed dispersers); leaf monkeys (L), *Ratufa* (R), *Callosciurus* (C), *Sundasciurus* (S) (all mainly seed predators). From Raemaekers *et al.*, 1980.

Family and genera	Species	Months with ripe fruit	*N*	*C*	*L*	Colour of ripe fruit externally	Eaten by
a. Euphorbiaceae							
Aporusa	sp.	Apr. 1976	1	L	17	Red aril	B, P, SB
Baccaurea	*racemosa*	Jun., Jul. 1976	2	L	15	Blue aril	
'Panchau'		Sept. 1976	1	M	15	Red aril	R, S
Sapium	*baccatum*	Oct. 1976	1	M	10		
Austrobuxus	*nitidus*	Mar. 1977	1	U	15	Red aril	
b. Burseraceae							
Dacryodes	*rugosa*	Jul., Aug., 1976	4	L	17	purplish	3HB, P
Dacryodes	*rostrata*	Nov. 1976	1	U	28	purplish	B, SB
Santiria	*laeviga*	Feb., Mar. 1977	3	U	15	purplish	
Santiria	*rubiginosa*	Apr. 1977	1	M	11	purplish	R, C, S
Santiria	*tomentosa*	May 1977	1	U	19	purplish	
c. Myristicaceae							
Horsfieldia	*superba*	May 1976	1	U	?	yellow	
Knema	sp. A	Jun. 1976	1	M	45	yellow	HB, M
Horsfieldia	*sucosa*	Aug. 1976	2	U	55	pink	
Horsfieldia	(?) *macroma*	Nov. 1976	2	U	?	yellow	L, C
Knema	sp. B	Dec. 1976	1	M	40	yellow	
Myristica	sp.	Feb. 1977	2	M	65	yellowish	
Knema	sp. A	Jun. 1977	1	M	45	yellow	
d. Lauraceae							
(genera	A	Dec. 1976	1	U	27	dark green	2HB
unknown)	B	Feb. 1977	1	M	30	black	
	C	Jun. 1977	2	U	28	dark green	L, R
	D	Aug. 1977	1	U	30	pale green, white spots	
e. Myrtaceae and Guttiferae							
Eugenia	A	Jun. 1976	1	U	?	green	G, M, B
Eugenia	B	Jul. 1976	1	L	25	green	SS, P
Eugenia	C	Jul., Aug. 1976	4	U	25	green	
Eugenia	D	Sept. 1976	1	U	20	green	R, S
Eugenia	E	Nov., Dec. 1976	1	M	47	green	
Eugenia	F	Jan. 1977	1	U	14	black	
Calophyllum	*floribundum*	Apr. 1977	1	M	?	green	
Eugenia	G	May 1977	1	M	?	green	
Calophyllum	*curtisii*	Jun. 1977	4	U	10	greenish	
f. Leguminosae							
Dialium	*procerum*	Oct. 1976	2	U/E	44	black	G, M
Dialium	*platysepalum*	Oct., Nov. 1976	5	U/E	30	brown	
Dialium	*patens*	Dec. 1976	2	U/E	25	brown	L, R, C
Dialium	*laurinum*	Feb. 1977	1	U/E	?	brown	
g. Anacardiaceae							
Mangifera	*quadrifolia*	Apr. 1976	1	U	100	dull green	G, M
Mangifera	'temuor'	Jul. 1976	1	U	100	dark green	
Mangifera	'rawa'	Apr. 1977	1	M	33	pale green	R, C
Mangifera	'temuor'	Aug. 1977	1	U	100	dark green	

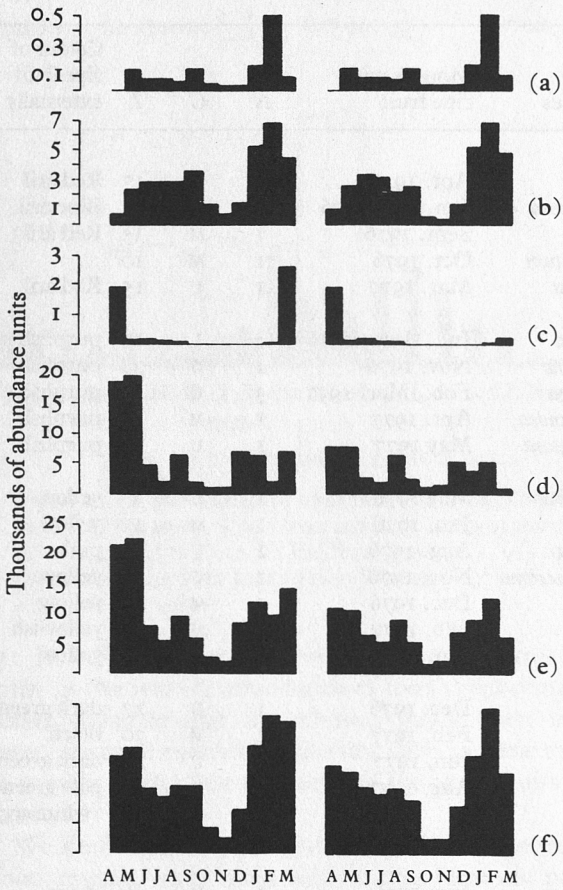


Figure 24.11. Variation between months in relative food abundance. No units are given on the scale of combined and weighted food abundance, because the weighting procedure renders the original units inapplicable. a, figs; b, other fruit; c, flowers; d, young leaves; e, combined; f, combined and weighted. (From Raemaekers, 1980.)

but do they use dry evergreen patches as bases from which to make sorties? Must every territory include a certain proportion of dry evergreen forest to be viable? Gittins and Tilson (Chapter 23) show how hoolock gibbons are restricted to good-quality natural forest (wetter semi-deciduous and evergreen), where they find sufficient fruit in a relatively small area.

If food limits gibbon populations, then their densities should be affected by the presence of competing species. Yet again, the numerous field studies do not test for this; in high diversity forest, however, it is not always clear who are competitors, since different species (especially more closely-related ones) eat different things for very variable lengths of time, e.g. all the time or rarely. Variation in forest types and recent history of human

disturbance can account for some or even all of the variation in gibbon density, irrespective of other species present. It cannot even be shown that congeneric gibbon species affect each other's densities in sympatry (Raemaekers, Chapter 19), although there are some indications that this may occur (Chivers, 1974; Marsh and Wilson, 1981). Furthermore, the large gibbon ranges at Kuala Lompat coincide with high densities of four monkey species with partial dietary overlap (MacKinnon and MacKinnon, 1978), whereas at Khao Yai, Thailand (evergreen forest of lower diversity), where there is only one other primate, gibbon home ranges are the smallest recorded outside Siberut (Table 24.4). Thus, a study in marginal habitat, with lower floristic diversity and more marked seasonality, would provide better opportunities for elucidating foraging strategies and a more striking contrast with existing optimal sites.

Table 24.4. Territory sizes of *Hylobates lar* in Khao Yai National Park, Thailand. Territory is defined as the area of exclusive use. Figures for groups other than A are minimum and maximum estimates (Raemaekers and Raemaekers, in prep.).

Group	Area (ha)
A	14
B	16-20
E	13-18
F	11-16
N	15-20

To conclude: we are obtaining a substantial ecological basis, in terms of population structure and density, diet, feeding and ranging behaviour, and habitat type and phenology, for helping to explain (and describe) the social structure that is unique to gibbons – monogamy and territoriality with elaborate songs.

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